

Midterm Assessment Answer Sheet

Semester 2, 2022/2023

Time allowed: 2 hours

Instructions (please read carefully):

- Write down your **Student Number** on the right and using ink or pencil, **shade completely** the corresponding bubbles in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
- This answer booklet comprises **NINETEEN (19) pages**, including this cover page.
- This is a **OPEN-SHEET** assessment. You are allowed one A4-sized double-sided cheatsheet.
- Weightage of questions is given in square brackets. The maximum attainable score is 100.
- You are allowed to bring a calculator, but it cannot have any form of external communication capability, i.e. not Wifi- or 4G-enabled. Mobile phones and tablets are not allowed.
- All questions must be answered in the space provided on the answer sheet; no extra sheets will be accepted as answers.
- You are allowed to write with pencils, as long as it is legible.
- Marks may be deducted** for unrecognisable handwriting and/or for not shading the student number properly.
- You must submit only the **ANSWER SHEET** and no other documents. The question set may be used as scratch paper.
- An excerpt of the question may be provided to aid you in answering in the correct box, where applicable. It is not the exact question. You should still refer to the original question in the question booklet.

STUDENT NUMBER														
A														
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For Examiner's Use Only

Question	Marks
Q1	/ 30
Q2	/ 28
Q3	/ 24
Q4	/ 15
Q5	/ 3
Total	/100

Question 1A State Representation

[2 marks]

Question 1B Initial and goal states

[2 marks]

Question 1C Which of the following statement(s) is/are true?

[3 marks]

- ☐ The search tree is finite.
- ☐ There are possibly many repeated states.
- ☐ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

Question 1D Best search algorithm. Explain.

[2 marks]

Question 1E State Representation for Variant #1

[2 marks]

Question 1F Initial and goal states for Variant #1

[2 marks]

Question 1G Invariant for Variant #1

[2 marks]

Question 1H Which of the following statement(s) is/are true for Variant #1? [3 marks]

- ☐ The search tree is finite.
- ☐ There are possibly many repeated states.
- ☐ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

Question 1I Best search algorithm for Variant #1. Explain.

[2 marks]

Question 1J To Graph-Search or not? That is the question.

[2 marks]

Question 1K State Representation for Variant #2

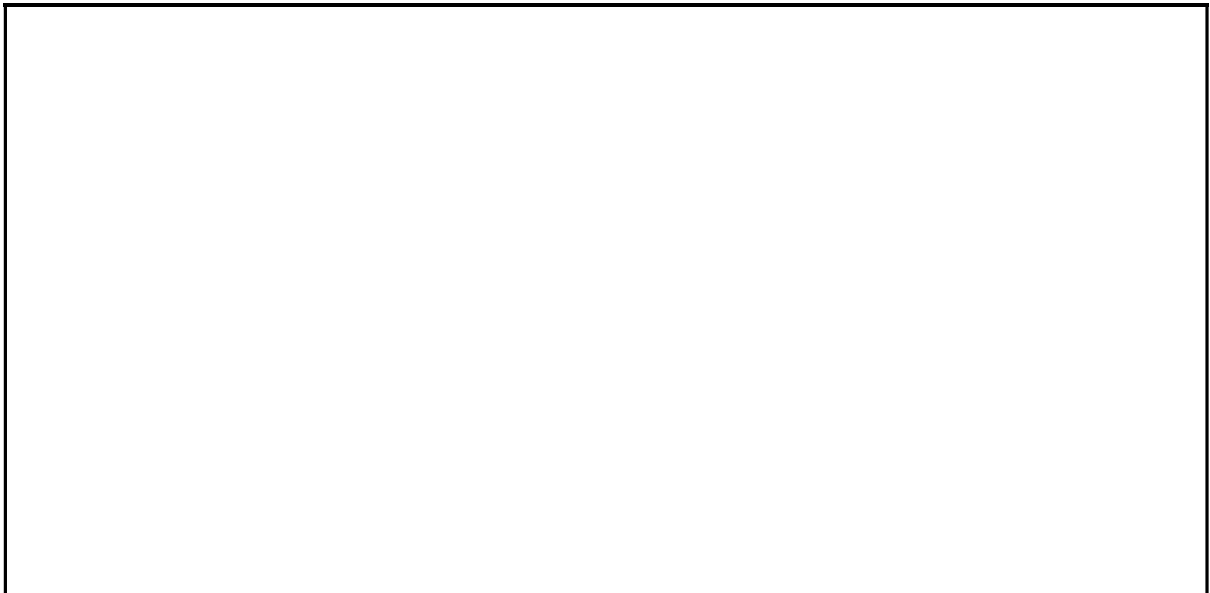
[2 marks]

**Question 1L** Which of the following statement(s) is/are true for Variant #2? [3 marks]

- ☐ The search tree is finite.
- ☐ There are possibly many repeated states.
- ☐ There are possibly multiple goal states.
- ☐ We can always find an answer (valid sequence of moves) if we employ the right search algorithm.
- ☐ None of the above.

Question 1M Best search algorithm for Variant #2. Explain.

[3 marks]



Question 2A Is h_1 admissible? Explain.

[2 marks]

☐ Admissible ☐ Not admissible**Question 2B** Is h_2 admissible? Explain.

[2 marks]

☐ Admissible ☐ Not admissible**Question 2C** Is h_A admissible? Explain.

[2 marks]

☐ Admissible ☐ Not admissible**Question 2D** h_B admissible and consistent? Explain.

[6 marks]

☐ Admissible ☐ Not admissible☐ Consistent ☐ Not consistent

Question 2E Is h_C admissible and consistent? Explain.

[6 marks]

☐ Admissible ☐ Not admissible

☐ Consistent ☐ Not consistent

Question 2F h_D admissible and consistent? Explain.

[6 marks]

☐ Admissible ☐ Not admissible

☐ Consistent ☐ Not consistent

Question 2G Dominant relationships between the heuristics.

[4 marks]

Which heuristic(s) does h_A dominate? Shade all that apply/ies.

- ☐ h_B ☐ h_C ☐ h_D ☐ None

Which heuristic(s) does h_B dominates Shade all that apply/ies.

- ☐ h_A ☐ h_C ☐ h_D ☐ None

Which heuristic(s) does h_C dominate? Shade all that apply/ies.

- ☐ h_A ☐ h_B ☐ h_D ☐ None

Which heuristic(s) does h_D dominate? Shade all that apply/ies.

- ☐ h_A ☐ h_B ☐ h_C ☐ None

Question 3A The price of a 2-room house?

[2 marks]

- ☐ High ☐ Low ☐ Undefined

Question 3B What is the entropy?

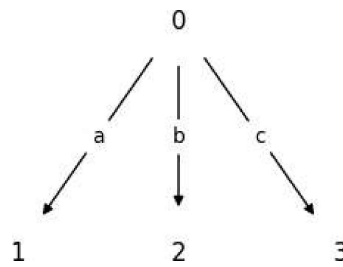
[2 marks]

- ☐ $0.0 \leq \text{Entropy} < 0.2$ ☐ $0.2 \leq \text{Entropy} < 0.4$
☐ $0.4 \leq \text{Entropy} < 0.6$ ☐ $0.6 \leq \text{Entropy} < 0.8$
☐ $0.8 \leq \text{Entropy} \leq 1.0$

Show your working below (if you wish):

Question 3C One-level Tree.

[6 marks]



What is node 0?

- | | | | |
|----------------------------|---------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☐ - |

What is node 1?

- | | | | |
|----------------------------|---------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☐ - |

What is node 2?

- | | | | |
|----------------------------|---------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☐ - |

What is node 3?

- | | | | |
|----------------------------|---------------------------------|--------------------------------|-----------------------------|
| ☐ Age | ☐ Condition | ☐ Location | ☐ Rooms |
| ☐ High | ☐ Low | ☐ Low/High | ☐ - |

What is edge a?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge b?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge c?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

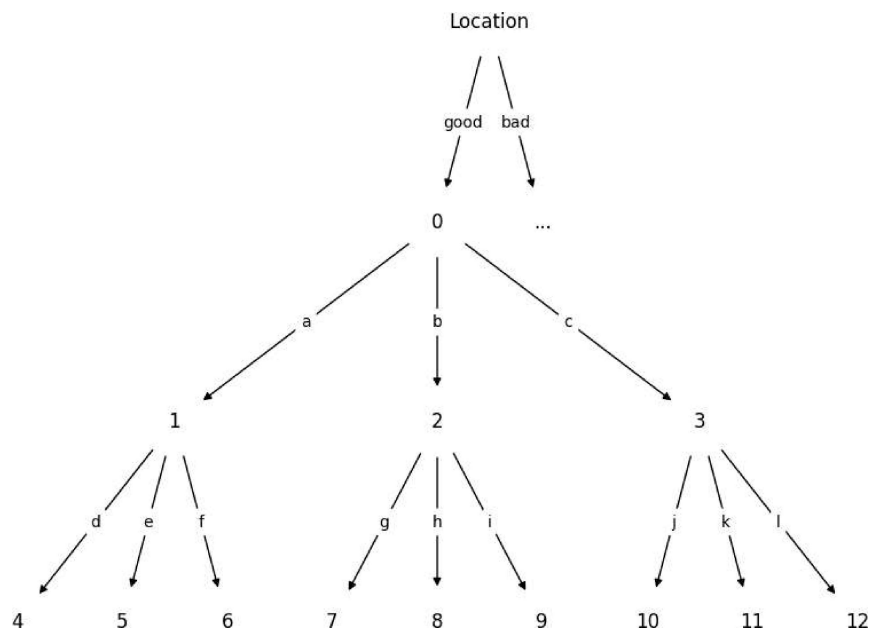
Question 3C One-level Tree. (Continued)

[6 marks]

Show your working below (if you wish):

Question 3D Full Decision Tree.

[6 marks]



What is node 0?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 1?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 2?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 3?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 4?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 5?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

What is node 6?

- | | | | |
|---------------------------|---------------------------------|-----------------------------|----------------------------|
| ☐ Age | ☐ Condition | ☐ Rooms | ☐ High |
| ☐ Low | ☐ Low/High | ☐ - | |

Are there any nodes from 7 onwards?

- | | |
|---------------------------|--------------------------|
| ☐ Yes | ☐ No |
|---------------------------|--------------------------|

Question 3D Full Decision Tree. (Continued)

[6 marks]

What is edge a?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

What is edge b?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

What is edge c?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

What is edge d?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

What is edge e?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

What is edge f?

☐ 2☐ mid☐ 3☐ new☐ aged☐ poor☐ excellent☐ -

Question 3D Full Decision Tree. (Continued)

[6 marks]

Show your working below (if you wish):

Question 3E Precision, Recall, F1.

[5 marks]

Number of True Positives (TP)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of False Positives (FP)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of True Negatives(TN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Number of False Negatives (FN)?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ >5

Precision?

- ☐ $0.0 \leq \text{Precision} < 0.2$ ☐ $0.2 \leq \text{Precision} < 0.4$
☐ $0.4 \leq \text{Precision} < 0.6$ ☐ $0.6 \leq \text{Precision} < 0.8$
☐ $0.8 \leq \text{Precision} \leq 1.0$

Recall?

- ☐ $0.0 \leq \text{Recall} < 0.2$ ☐ $0.2 \leq \text{Recall} < 0.4$
☐ $0.4 \leq \text{Recall} < 0.6$ ☐ $0.6 \leq \text{Recall} < 0.8$
☐ $0.8 \leq \text{Recall} \leq 1.0$

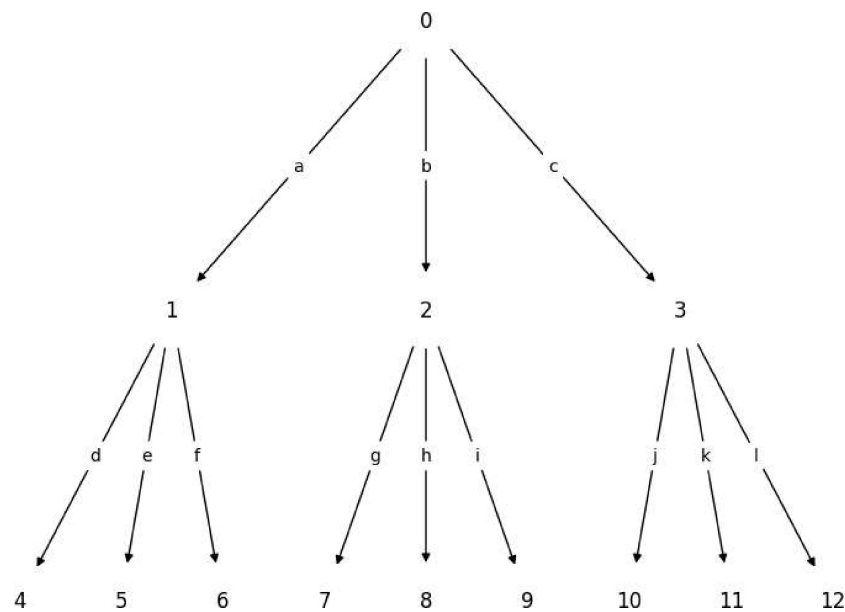
F1 Score?

- ☐ $0.0 \leq \text{F1 Score} < 0.2$ ☐ $0.2 \leq \text{F1 Score} < 0.4$
☐ $0.4 \leq \text{F1 Score} < 0.6$ ☐ $0.6 \leq \text{F1 Score} < 0.8$
☐ $0.8 \leq \text{F1 Score} \leq 1.0$

Show your working below (if you wish):

Question 3F Min-Sample DT.

[3 marks]



What is node 0?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 1?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 2?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 3?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 4?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 5?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

What is node 6?

- ☐ Age
☐ High

- ☐ Condition
☐ Low

- ☐ Location
☐ Low/High

- ☐ Rooms
☐ -

Are there any nodes from 7 onwards?

- ☐ Yes

- ☐ No

Question 3F Min-Sample DT. (Continued)

[3 marks]

What is edge a?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge b?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge c?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge d?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge e?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

What is edge f?

- | | | | | |
|----------------------------|---------------------------|----------------------------|----------------------------|---------------------------------|
| ☐ 2 | ☐ 3 | ☐ aged | ☐ bad | ☐ excellent |
| ☐ good | ☐ mid | ☐ new | ☐ poor | ☐ - |

Show your working below (if you wish):

Question 4A Game Tree.

[6 marks]

What is the minimum number of variables required?

- ☐ 1 ☐ 2 ☐ 3 ☐ 7 ☐ 8 ☐ Other

How many nodes (not including the leaves) are there in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

What is the number of leaves where the first player wins?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the number of leaves where the first player loses?

- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ >6

What is the total number of edges in the game tree?

- ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☐ 8 ☐ 9 ☐ 10 ☐ 11 ☐ 12 ☐ 13 ☐ 14
☐ 15 ☐ 16 ☐ 17 ☐ 18 ☐ 19 ☐ 20 ☐ >20

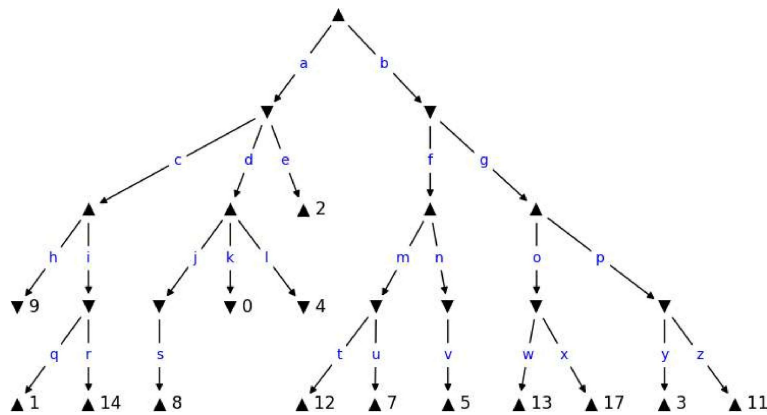
Which player do we expect to win?

- ☐ First player ☐ Second player ☐ It depends ☐ Draw

Show your working below (if you wish):

Question 4B Alpha-beta left-to-right.

[5 marks]



What is the root value?

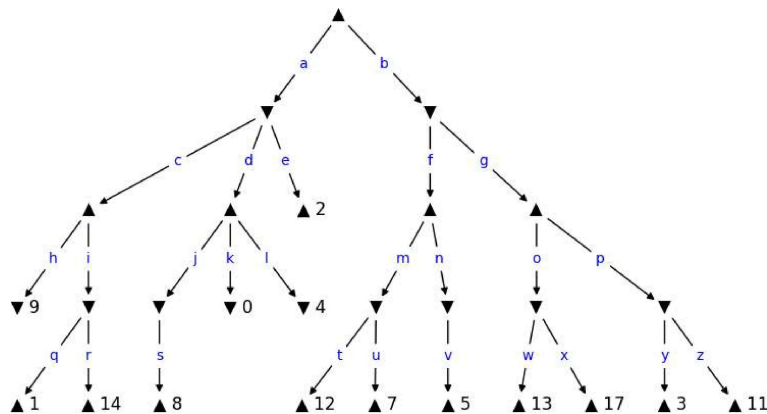
- ☐ 0 ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 7
☐ 8 ☐ 9 ☐ 11 ☐ 12 ☐ 13 ☐ 14 ☐ 17

Which of the following link(s) are pruned? Shade all that is/are true.

- | | | | | | |
|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| ☐ a | ☐ b | ☐ c | ☐ d | ☐ e | ☐ f |
| ☐ g | ☐ h | ☐ i | ☐ j | ☐ k | ☐ l |
| ☐ m | ☐ n | ☐ o | ☐ p | ☐ q | ☐ r |
| ☐ s | ☐ t | ☐ u | ☐ v | ☐ w | ☐ x |
| ☐ y | ☐ z | | | | |

Question 4C Alpha-beta right-to-left.

[4 marks]

Which of the following link(s) are pruned? Shade all that is/are true.

- | | | | | | |
|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| ☐ a | ☐ b | ☐ c | ☐ d | ☐ e | ☐ f |
| ☐ g | ☐ h | ☐ i | ☐ j | ☐ k | ☐ l |
| ☐ m | ☐ n | ☐ o | ☐ p | ☐ q | ☐ r |
| ☐ s | ☐ t | ☐ u | ☐ v | ☐ w | ☐ x |
| ☐ y | ☐ z | | | | |

Question 5 3 most important lessons

[3 marks]

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It may be used as scratch paper.

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